

Applying fuzzy picture public encryption in corporate security models

Instructor Assoc.Prof. Hai Pham Van¹,
Student Vu Nguyen Anh Tuan², Student Tran Quang Thang³,
Student Tran Nguyen Phuc⁴, Student Nguyen Phan Vi⁵,
Student Phan Cong Truong Son⁶

¹School of Information Technology and Communication, Hanoi
University of Science and Technology, Hai Ba Trung, Hanoi, Vietnam.

²Hanoi University of Science and Technology, Hai Ba Trung, Hanoi,
Vietnam.

Abstract

This report presents an advanced approach to implementing corporate security models by utilizing Picture Fuzzy Public Key Encryption (PFPKE), specifically leveraging biometric data for secret key generation. Traditional public key encryption models are often incompatible with biometrics because such data is inherently prone to interference, blurriness, and slight variations during each collection process. To address this, the study introduces the concept of Picture Fuzzy Public Key Encryption (PFPKE), a model designed to accept partial fuzzy data - essentially an interfered version of the original biometric signature - as a valid private key for decryption. Unlike conventional systems where private keys must be stored on physical hardware like USB devices, the PFPKE model eliminates the need for external storage equipment, providing a more seamless and integrated security solution for modern enterprises.

Keywords: Picture fuzzy public key, fuzzy data, biometric, corporate security, PFPKE

1 Introduction

In conventional corporate public-key systems, securing an employee's private key is critical to maintaining system integrity. The standard industry solution is to store the

private key on a physical device (like a smart card or USB) and require the user to memorize an activation password.

Using biometric data (such as fingerprints or irises) is a superior alternative because it is entirely unique to each individual, offering both maximum convenience and safety.

Because biometric readings fluctuate and can be interfered with each time they are scanned, they cannot be directly applied to traditional encryption schemes that demand exact mathematical matches.

The paper introduces the concept of a "fuzzy signature," which allows biometrics to generate secret keys independently. This breakthrough enables the practical application of *fuzzy picture public encryption* (PFPKE) in corporate biometric security models.

2 Related Works

In this section, we collected some other works relating to fuzzy public keys, biometric security, and advanced key generation frameworks. The foundational approach for this study is presented in the report [1], which introduces the implementation of security models using users' biometric data to make secret keys.

There are many similar works about the topic of biometric-based key generation. For example, there is a work about fuzzy extractors for biometric identification [2], a work proposes a new method for generating an invariant iris private key based on the fuzzy vault system [3], and another improved method like the fuzzy commitment scheme for the generation of cryptographic keys based on iris biometrics [4].

Over the years, researchers have created tools for ensuring the robustness and reusability of these cryptographic keys. This includes work on robust and reusable fuzzy extractors for low-entropy distributions and their application to user authentication [5], as well as research into information-theoretically secure, robust, and reusable fuzzy extractors designed for structured sources [6].

We also include research about the application of artificial intelligence and machine learning in security. For instance, there is work regarding neural fuzzy extractors as a secure way to use artificial neural networks for biometric user authentication [7]. This development has evolved into more recent methods, such as cryptographic key generation using deep learning with biometric face and finger vein data [8].

Therefore, we introduced more complex frameworks such as biometrics-based authenticated key exchange with a multi-factor fuzzy extractor [9]. Some implementations like a fuzzy logic-based cryptographic framework for real-time dynamic key generation [10] help improve data encryption. There are also papers to analyze other methods for reducing authentication vulnerabilities, such as revisiting fuzzy signatures to move toward a more risk-free cryptographic authentication system based on biometrics [11].

3 Proposed Model

To handle the PFPKE framework described in the report, we need an AI model to function as a "fuzzy picture extraction algorithm" and it must be capable of processing

high-dimensional, "noisy" biometric data and mapping it into the specific Picture Fuzzy Set (PFS) membership degrees (u,n,v) required for the cryptographic keys.

3.1 Deep Picture Fuzzy Neural Network (DPFNN)

Based on the requirements, we have concluded that a Deep Picture Fuzzy Neural Network (DPFNN) is the most suitable architecture for this framework. Unlike standard neural networks that output binary or simple fuzzy values, a DPFNN uses a specialized output layer to satisfy the Picture Fuzzy constraint:

$$\mu_A(x) + \nu_A(x) + \pi_A(x) < 1 \quad \forall x \in X$$

3.2 Model Architecture

- **Feature Extraction Layer (CNN/Transformer):** If the biometric source is an image (e.g., iris or fingerprint), a Convolutional Neural Network (CNN) extracts spatial features to form the vector x . Deep layers are employed to learn invariant representations, ensuring that even if the input is blurred or noisy, the essential identity features are preserved. The output is a normalized feature vector x , which serves as the center for subsequent fuzzy computations.
- **Fuzzy Membership Layer:** A custom layer applies the L-R fuzzy model and triangular fuzzy functions to compute:
 - u : Degree of certainty in the match
 - n : Degree of ambiguity or noise
 - v : Degree of certainty of mismatch
- **Optimization Function:** The model is trained to minimize the False Rejection Rate (FRR) and False Acceptance Rate (FAR). This is achieved by adaptively adjusting the threshold value t based on the security parameter λ .

4 Experimental Results

The report confirms that the proposed framework is mathematically sound and satisfies the following security criteria:

- **Security Threshold (t):** The False Acceptance Rate (FAR) is negligible with respect to the security parameter λ . This implies that the probability of an unauthorized user being incorrectly accepted is extremely low.
- **Error Tolerance:** For all legitimate biometric variations x' , the False Rejection Rate (FRR) remains below a predefined error bound.
- **Decryption Accuracy:** The system successfully recovers the message M if the distance between the enrollment vector x and the query vector x' satisfies:

$$d(x, x') \leq t$$

- **IND-CCA Security:** The report demonstrates that the proposed PFPKE scheme achieves indistinguishability under chosen-ciphertext attacks (IND-CCA), providing strong resistance against modern cryptanalytic adversaries.

4.1 Extraction Results

The algorithm first processes raw biometric input into a standardized representation:

- **Biometric Vector:** The user's biometric data is transformed into a feature vector, denoted as x .
- **Fuzzy Output:** The extraction algorithm maps x into a Picture Fuzzy Set (PFS) representation:
 - Positive degree: 0.75
 - Neutral degree: 0.10
 - Negative degree: 0.15

4.2 Key Generation and Setup

The experiment employs a high-security random variable for the security parameter λ .

- **Public Parameters:** The setup algorithm generates a Base64-encoded string:
MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBABJQVu6lHAEtia3xc8fCEkd9dpt0jGt7FSiMz
- **Public Key:** The KeyGen algorithm produces a composite public key consisting of multiple cryptographic components.

4.3 Encryption and Decryption Performance

The system is evaluated using a sample plaintext message:

- **Input Message (M):** "Fuzzy picture public encryption"
- **Ciphertext (CT):** The encryption process produces:

exhxWbrXSm7huDc/4LkAHnmk3i91K1FDBqUwPk01gLR0pY80w/Sqe3xPLpdkSpoTLwm/T3KqIOqGs8HehzXHHw==

- **Successful Recovery:** Upon re-acquiring biometric data (simulated as x'), the decryption algorithm successfully reconstructs the original message M .

4.4 Evaluation and Security Implications

The experimental results demonstrate the effectiveness of the proposed framework:

- **Verification:** The correctness condition holds:

$$\text{Decode}(\text{par}, pk_f, x', CT) = M$$

even when using noisy or fuzzy biometric inputs.

- **Practicality:** A single biometric vector is sufficient to generate a valid IND-CCA secure ciphertext, eliminating the need for external storage such as USB tokens or passwords.

- **Security Guardrail:** The neutral degree ($n = 0.10$) represents uncertain regions in the biometric data and must be carefully monitored to prevent potential exploitation.

5 Conclusion

Traditionally, public-key cryptographic schemes require users to store their private keys on physical devices such as USB tokens to ensure security. However, the need to carry such hardware continuously is often impractical in real-world applications.

Biometric data has emerged as a promising alternative for key generation due to its uniqueness and convenience. Nevertheless, the inherent variability and noise present in biometric acquisition make it unsuitable for direct use in conventional encryption schemes.

To address this limitation, this paper introduces a novel framework based on *fuzzy picture public encryption* (FPFPE), in which partial or imprecise biometric data can be utilized as a private key for decryption without requiring any auxiliary information. This approach eliminates the need for traditional password-based or hardware-based authentication mechanisms.

Despite these advantages, a critical aspect of the proposed model lies in the management of the neutral degree within the fuzzy set representation. Specifically, the neutral degree corresponds to uncertain or indeterminate regions in the biometric data. If not properly controlled, these regions may introduce potential vulnerabilities that could be exploited by adversaries to gain unauthorized access.

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